DataBase Lab 07 (POST-Lab Task)

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Task 1

Welcome Page | hr

Worksheet | Query Builder

```
SELECT DISTINCT e.employee_id, e.first_name
FROM employees e
JOIN job_history j
ON e.employee_id = j.employee_id;
```

Script Output | Query Result

SQL | All Rows Fetched: 10 in 0.002 seconds

	EMPLOYEE_ID	FIRST_NAME
1	104	Bruce
2	114	Den
3	201	Michael
4	102	Lex
5	101	Neena
6	105	David
7	103	Alexander
8	122	Payam
9	200	Jennifer
10	176	Jonathon

Task 2

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Worksheet | Query Builder

```
SELECT e.first_name, e.department_id
FROM employees e
JOIN employees m
ON e.manager_id = m.employee_id
WHERE e.department_id = m.department_id;
```

Script Output | Query Result

SQL | Fetched 50 rows in 0.002 seconds

	FIRST_NAME	DEPARTMENT_ID
1	Lex	90
2	Neena	90
3	Diana	60
4	Valli	60
5	David	60
6	Bruce	60
7	Luis	100
8	Jose Manuel	100
9	Ismael	100
10	John	100
11	Daniel	100
12	Karen	30
13	Guy	30
14	Sigal	30
15	Shelli	30
16	Alexander	30
17	Girard	50
18	Martha	50
19	Jean	50
20	Winston	50
21	Steven	50
22	James	50
23	Irene	50
24	Julia	50
25	Anthony	50
26	Julia	50
27	Alexis	50
28	Nandita	50
29	TJ	50
30	James	50

Task 3

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Worksheet | Query Builder

```sql
SELECT first_name, department_id
FROM employees
WHERE department_id IN
(
    SELECT department_id
    FROM employees
    GROUP BY department_id
    HAVING COUNT(*) > 5
);
```

Script Output | Query Result

SQL | Fetched 50 rows in 0.004 seconds

	FIRST_NAME	DEPARTMENT_ID
1	Alexander	60
2	Valli	60
3	Shelli	30
4	Irene	50
5	James	50
6	Steven	50
7	Laura	50
8	Hazel	50
9	Curtis	50
10	Alberto	80
11	Oliver	80
12	Lindsey	80
13	Clara	80
14	Mattea	80
15	Tayler	80
16	Martha	50
17	Kelly	50
18	Timothy	50
19	Britney	50
20	Alexander	30
21	Kevin	50
22	Jason	50
23	Peter	50
24	Peter	80
25	Patrick	80
26	Danielle	80
27	Amit	80
28	Elizabeth	80

Task 4

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Worksheet | Query Builder

```
SELECT e.first_name AS employee,
       m.first_name AS manager
FROM employees e
LEFT JOIN employees m
ON e.manager_id = m.employee_id;
```

Script Output | Query Result

SQL | All Rows Fetched: 109 in 0.006 seconds

	EMPLOYEE	MANAGER
1	Sundita	Gerald
2	Elizabeth	Gerald
3	William	Gerald
4	Tayler	Gerald
5	Harrison	Gerald
6	Lisa	Gerald
7	Alexander	Lex
8	Amit	Alberto
9	Sundar	Alberto
10	David	Alberto
11	Mattea	Alberto
12	Danielle	Alberto
13	Clara	Alberto
14	Anthony	Adam
15	Julia	Adam
16	Alexis	Adam
17	Nandita	Adam
18	TJ	Adam
19	James	Adam
20	Mozhe	Adam
21	Laura	Adam
22	Luis	Nancy
23	Jose Manuel	Nancy
24	Ismael	Nancy
25	John	Nancy
26	Daniel	Nancy
27	Pat	Michael
28	William	Shelley
29	Diana	Alexander
30	Valli	Alexander
31	David	Alexander

Task 5

Welcome Page | hr

Worksheet | Query Builder

```
SELECT first_name, salary
FROM employees
WHERE salary >
      (SELECT AVG(salary)
       FROM employees e2
       WHERE e2.department_id = employees.department_id)
AND salary <
      (SELECT MAX(salary) FROM employees);
```

Script Output | Query Result

SQL | All Rows Fetched: 36 in 0.008 seconds

	FIRST_NAME	SALARY
1	Daniel	9000
2	Nancy	12008
3	Den	11000
4	Steven	24300
5	Michael	13000
6	Shelley	12008
7	Britney	3900
8	Sarah	4000
9	Jennifer	3600
10	Kelly	3800
11	Alexis	4100
12	Nandita	4200
13	Trenna	3500
14	Renske	3600
15	Kevin	5800
16	Shanta	6500
17	Payam	7900
18	Adam	8200
19	Matthew	8000
20	Ellen	11000
21	Tayler	9600
22	Harrison	10000
23	Lisa	11500
24	Danielle	9500
25	Clara	10500
26	Allan	9000
27	Patrick	9500
28	Janette	10000

Task 6

Welcome Page | hr

Worksheet | Query Builder

```
SELECT e.employee_id, e.first_name, j.department_id
FROM employees e
JOIN job_history j
ON e.employee_id = j.employee_id;
```

Script Output | Query Result

SQL | All Rows Fetched: 15 in 0.006 seconds

	EMPLOYEE_ID	FIRST_NAME	DEPARTMENT_ID
1	105	David	90
2	102	Lex	80
3	102	Lex	60
4	104	Bruce	20
5	201	Michael	20
6	103	Alexander	40
7	122	Payam	50
8	101	Neena	60
9	101	Neena	110
10	101	Neena	110
11	114	Den	50
12	176	Jonathon	80
13	176	Jonathon	80
14	200	Jennifer	90
15	200	Jennifer	90

Task 7

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Worksheet | Query Builder

```
SELECT DISTINCT e.employee_id, e.first_name
FROM employees e
JOIN job_history j
ON e.employee_id = j.employee_id
JOIN departments d
ON j.department_id = d.department_id
JOIN locations l
ON d.location_id = l.location_id
JOIN countries c
ON l.country_id = c.country_id
JOIN regions r
ON c.region_id = r.region_id;
```

Script Output | Query Result

SQL | All Rows Fetched: 10 in 0.009 seconds

	EMPLOYEE_ID	FIRST_NAME
1	104	Bruce
2	114	Den
3	201	Michael
4	102	Lex
5	101	Neena
6	105	David
7	103	Alexander
8	122	Payam
9	200	Jennifer
10	176	Jonathon

Task 8

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Worksheet | Query Builder

```
SELECT e1.first_name, e1.last_name, e1.department_id
FROM employees e1
JOIN employees e2
ON e1.last_name = e2.last_name
AND e1.department_id <> e2.department_id;
```

Script Output | Query Result

SQL | All Rows Fetched: 4 in 0.002 seconds

	FIRST_NAME	LAST_NAME	DEPARTMENT_ID
1	Steven	King	90
2	Janette	King	80
3	Winston	Taylor	50
4	Jonathon	Taylor	80

Task 9

Welcome Page | hr

Worksheet | Query Builder

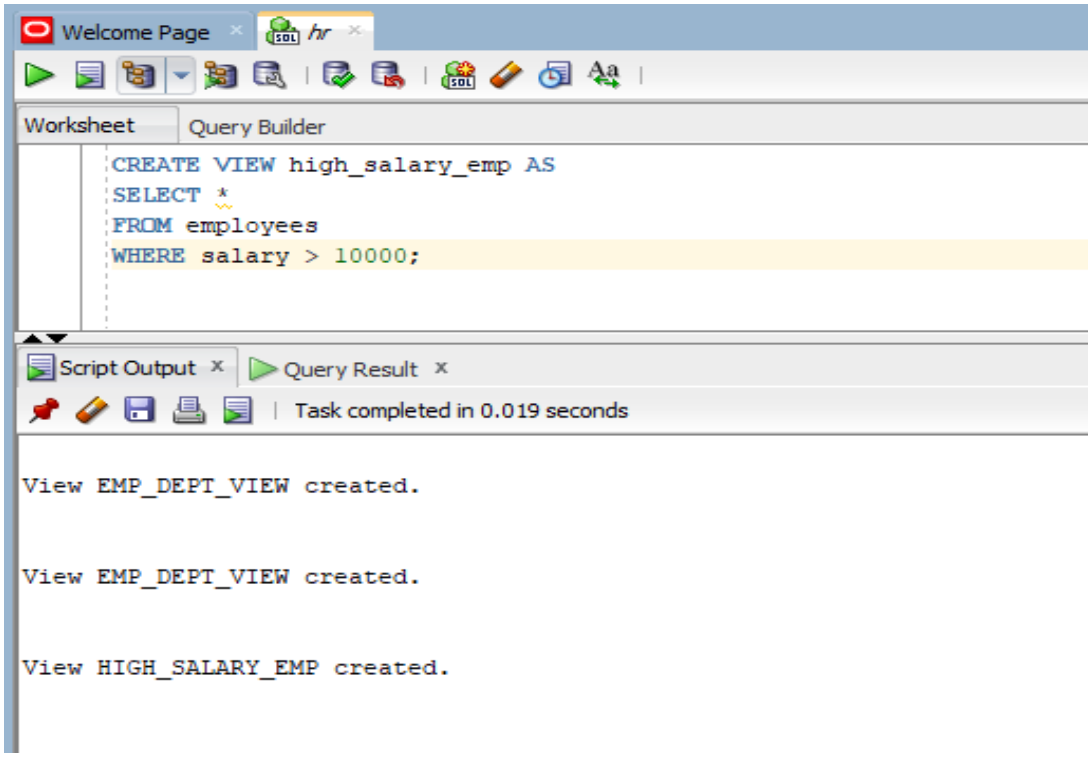
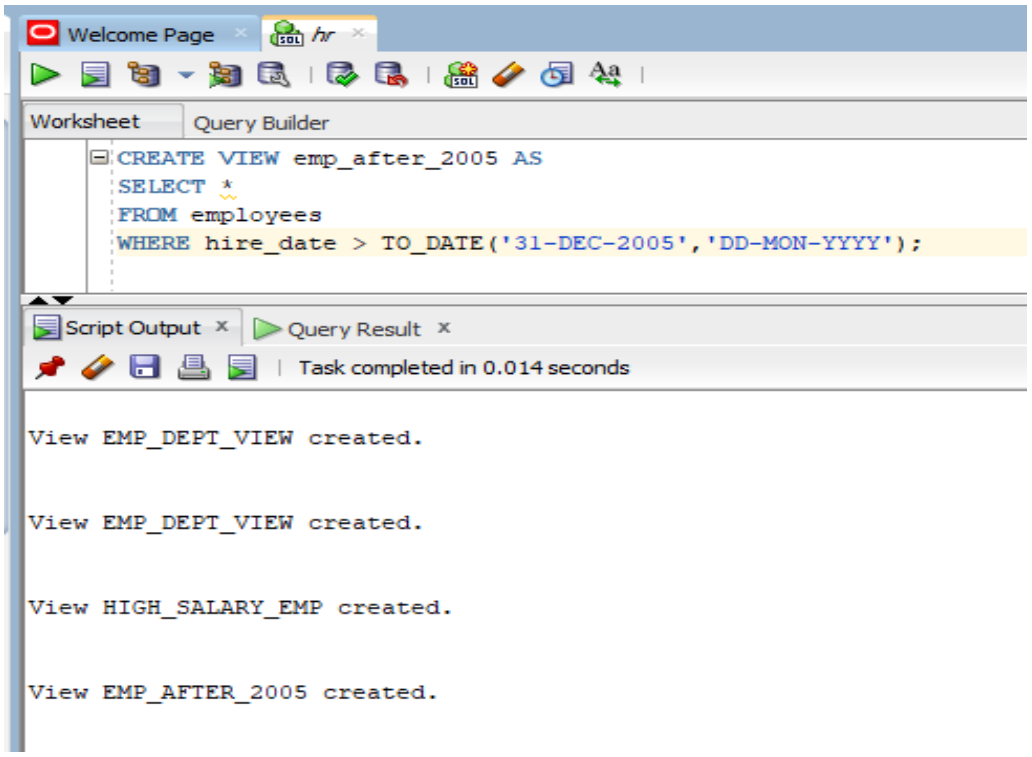
```
SELECT *
FROM (
    SELECT first_name, department_id, salary,
           DENSE_RANK() OVER
           (PARTITION BY department_id ORDER BY salary DESC) rnk
    FROM employees
)
WHERE rnk <= 3;
```

Script Output | Query Result

SQL | All Rows Fetched: 27 in 0.003 seconds

	FIRST_NAME	DEPARTMENT_ID	SALARY	RNK
1	Jennifer	10	4400	1
2	Michael	20	13000	1
3	Pat	20	6000	2
4	Den	30	11000	1
5	Alexander	30	3100	2
6	Shelli	30	2900	3
7	Susan	40	6500	1
8	Adam	50	8200	1
9	Matthew	50	8000	2
10	Payam	50	7900	3
11	Mujtaba	60	60000	1
12	Alexander	60	9000	2
13	Bruce	60	6000	3
14	Hermann	70	10000	1
15	John	80	14000	1
16	Karen	80	13500	2
17	Alberto	80	12000	3
18	Steven	90	24300	1
19	Neena	90	17000	2
20	Lex	90	17000	2
21	Nancy	100	12008	1
22	Daniel	100	9000	2
23	John	100	8200	3
24	Shelley	110	12008	1
25	William	110	8300	2
26	Sohail	(null)	(null)	1
27	Kimberely	(null)	7000	2

Task 10



Task 11